Fill in the table

Two strings of DNA letters. Where do they differ?

The reference is **GATTACA**. The read is **GATACA**. One letter is missing from the read. The table finds out which one.

The rule for one cell

Look at three neighbours and work out three numbers:

from the **upper-left** cell:	same letters: **+** 2 different letters: -1
from the cell **above**:	-2
from the cell to the **left** :	-2

Write the **biggest** of the three. If the biggest is below 0, write **0**.

The border, in italics, is all zeros. Two cells are done for you:

- G against G: upper-left is 0, same letters, so $0 + 2 = 2$. Above gives -2, left gives -2. Write 2.
- G against A: upper-left gives $0 - 1 = -1$, above gives -2, left gives $2 - 2 = 0$. Write 0.

		G	**A**	**T**	**T**	**A**	**C**	**A**
	0	*0*	*0*	*0*	*0*	*0*	*0*	*0*
G	*0*	2	0					
A	*0*							
T	*0*							
A	*0*							
C	*0*							
A	*0*							

When the table is full

1. What is the biggest number in the table, and where is it?
2. Start there. Ask which neighbour that number came from, and step to it. Repeat until you reach a 0. Draw the path.
3. One step of your path goes **left** instead of diagonally. That is the missing letter. Which one is it?
4. Which cells did you fill in that the path never came near?

Answer

		G	**A**	**T**	**T**	**A**	**C**	**A**
	0	*0*	*0*	*0*	*0*	*0*	*0*	*0*
G	*0*	[2]	0	0	0	0	0	0
A	*0*	0	[4]	[2]	0	2	0	2
T	*0*	0	2	6	[4]	2	1	0
A	*0*	0	2	4	5	[6]	4	3
C	*0*	0	0	2	3	4	[8]	6
A	*0*	0	2	0	1	5	6	[10]

1. The biggest number is **10**, in the bottom-right cell.
2. The path is the boxed cells: $10 \leftarrow 8 \leftarrow 6 \leftarrow 4 \leftarrow 2 \leftarrow 4 \leftarrow 2$.
3. The step from the boxed 4 in row T back to the boxed 2 in row A goes diagonally, and the step from that 2 back to the boxed 4 in row A goes **left**. So the read skips the first **T** of the reference:

```
reference:  G A T T A C A
read:       G A _ T A C A
```

"Attentions" Added

		T	**A**	**T**	**T**	**A**	**T**	**G**	**C**	**A**
	0	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*
T	*0*	2	0							
A	*0*									
T	*0*									
T	*0*									
T	*0*									
G	*0*									
C	*0*									
A	*0*									

Answer

		T	**A**	**T**	**T**	**A**	**T**	**G**	**C**	**A**
	0	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*
T	*0*	[2]	0	2						
A	*0*		[4]	2	1					
T	*0*			[6]	4	2				
T	*0*				[8]	[6]	4			
T	*0*					7	[8]	6		
G	*0*						6	[10]	8	
C	*0*							8	[12]	10
A	*0*								10	[14]

1. The biggest number is **14**, in the bottom-right cell.
2. The path is the boxed cells: 14 ← 12 ← 10 ← 8 ← 6 ← 8 ← 6 ← 4 ← 2.
3. The step from the boxed 8 in row T back to the boxed 6 in row T goes diagonally, and the step from that 6 back to the boxed 8 in row T goes **left**. So the read skips the second **A** of the reference:

```
reference:  T A T T A T G C A
read:       T A T T _ T G C A
```

4. 23 cells. The whole table has 72.